BEFORE YOU READ

Fourteen limits in five families

The special limits on this page group by the functions involved and by where x is heading.

- **L0 Special limits**
  - **L1 Trigonometric, $x \to 0$**
    - L2 $\sin x / x$ — 1
    - L2 $(1 - \cos x) / x$ — 0
    - L2 $(1 - \cos x) / x^2$ — $1/2$
    - L2 $\tan x / x$ — 1
    - L2 $x / \sin x$ — 1
  - **L1 Exponential, $x \to 0$**
    - L2 $(e^x - 1) / x$ — 1
    - L2 $(a^x - 1) / x$ $a > 0$ — $\ln a$
    - L2 $\ln(1 + x) / x$ — 1
  - **L1 Definition of e**
    - L2 $(1 + 1/x)^x$ as $x \to \infty$ equivalently $(1 + x)^{1/x}$ as $x \to 0$ — e
  - **L1 Logarithmic boundary, $x \to 0^+$**
    - L2 $x \ln x$ — 0
    - L2 $x^n \ln x$ $n > 0$ — 0
  - **L1 Growth rate, $x \to \infty$**
    - L2 $\ln x / x^n$ $n > 0$ — 0
    - L2 x^n / e^x any n — 0
    - L2 x^n / a^x any n, $a > 1$ — 0

TAKEAWAY

Two directions

The trigonometric and exponential limits look at $x \to 0$; the growth-rate limits look at $x \to \infty$ and rank logarithms below powers below exponentials.